



ShopNest E-Commerce Performance & Delivery Analysis

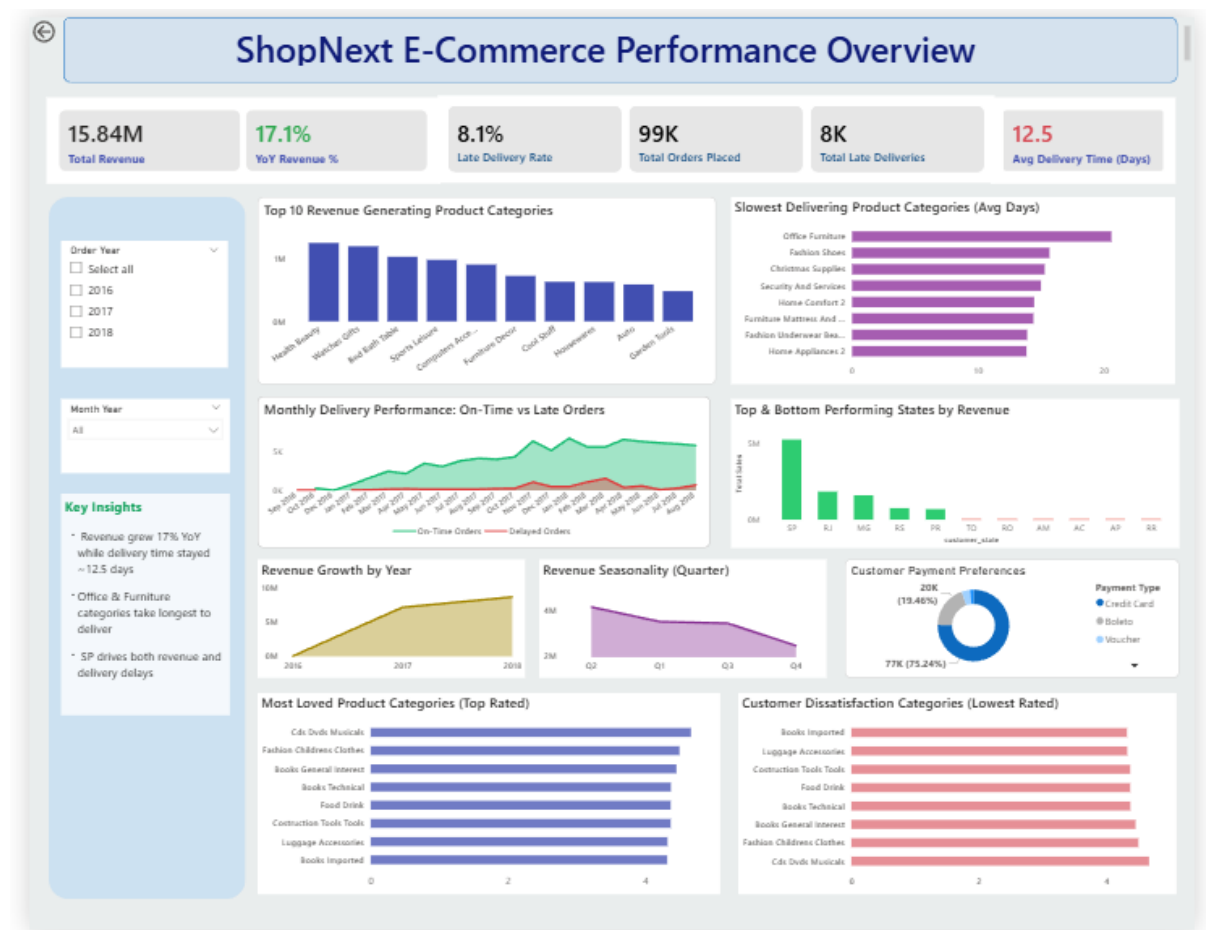
Power BI Capstone Project Report

1 Executive Overview

Question Statement

How is ShopNest performing overall in terms of revenue growth, order volume, and delivery reliability?

Visualization



Explanation

The executive dashboard provides a high-level view of business performance between 2016 and 2018.

Key Metrics:

- **Total Revenue:** 15.84M
- **YoY Revenue Growth:** 17.1%

- **Total Orders:** 99K
- **Late Delivery Rate:** 8.1%
- **Total Late Deliveries:** 8K
- **Average Delivery Time:** 12.5 Days

Key Insights:

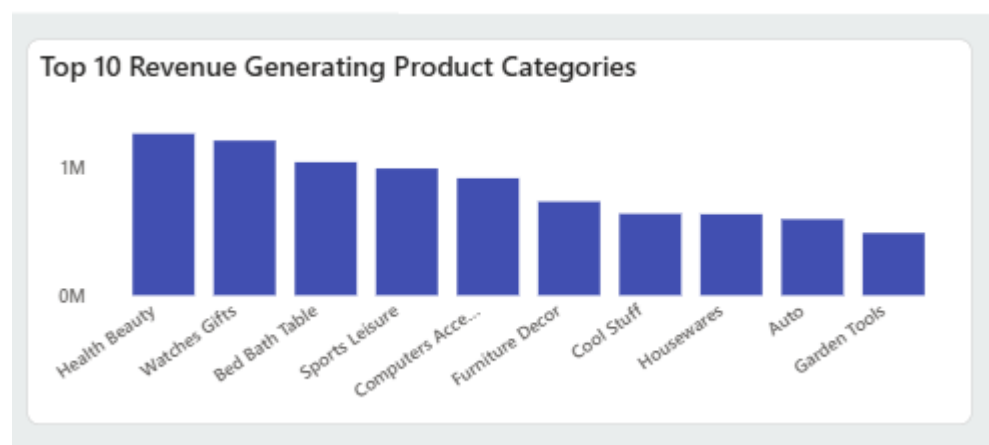
- Revenue is growing steadily at 17% YoY.
- Despite growth, delivery time remains relatively high at 12.5 days.
- Only 8.1% of deliveries are late, indicating strong SLA compliance.
- However, delivery duration suggests operational optimization opportunities.

2 Revenue Drivers

Question Statement

Which product categories contribute the most to revenue?

Visualization



Explanation

The chart identifies the top revenue-generating product categories.

Key Findings:

- Health & Beauty, Watches & Gifts, and Bed & Bath are leading revenue contributors.
- Revenue concentration is skewed toward lifestyle and consumer goods categories.
- High-performing categories may require supply chain prioritization to sustain growth.

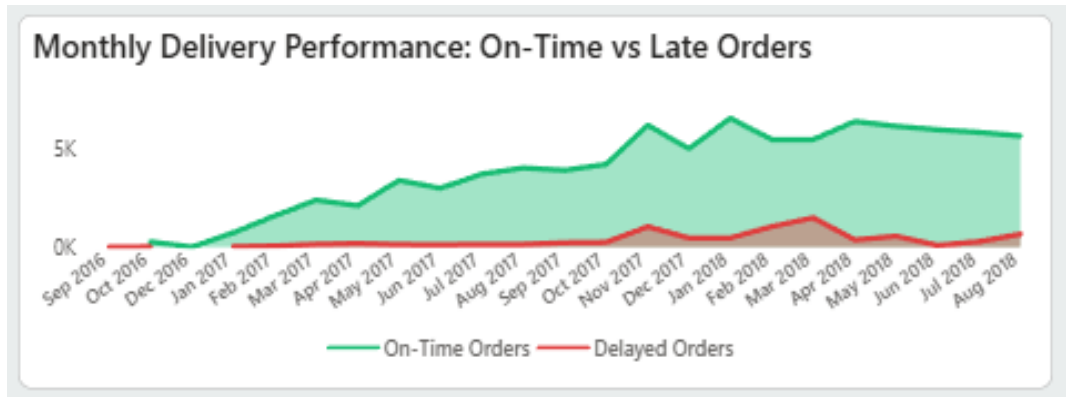
This insight supports product-level strategy and inventory planning.

3 Delivery Performance Over Time

Question Statement

How has delivery performance evolved monthly, and are late deliveries increasing?

Visualization



Explanation

The monthly trend chart compares on-time vs late deliveries over time.

Observations:

- On-time deliveries consistently exceed delayed orders.
- Order volume increases over time, aligned with revenue growth.
- Slight increase in delayed deliveries observed during peak sales months.

This suggests operational strain during high-demand periods.

📌 Slowest Delivering Categories

Question Statement

Which product categories take the longest time to deliver?

Visualization



Explanation

This chart ranks categories by **Average Delivery Time (Days)**.

Key Insights:

- Office Furniture and Furniture-related categories have the longest delivery times.
- Bulky items correlate with longer fulfillment cycles.
- Lightweight consumer goods show shorter delivery durations.

Operational Recommendation:

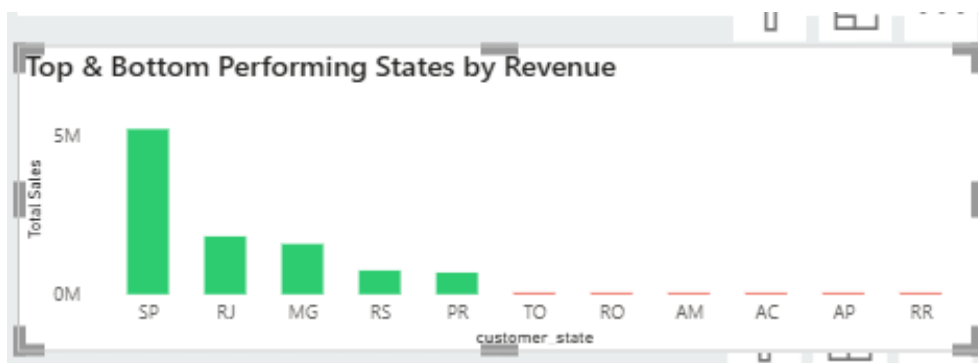
- Optimize logistics partnerships for heavy goods.
- Introduce regional warehousing for bulky categories.

5 Geographic Revenue Performance

Question Statement

Which states generate the highest and lowest revenue?

Visualization



Explanation

Revenue is heavily concentrated in:

- São Paulo (SP) – highest contributor
- Followed by RJ and MG

Lower-performing states include northern and smaller regions.

Insight:

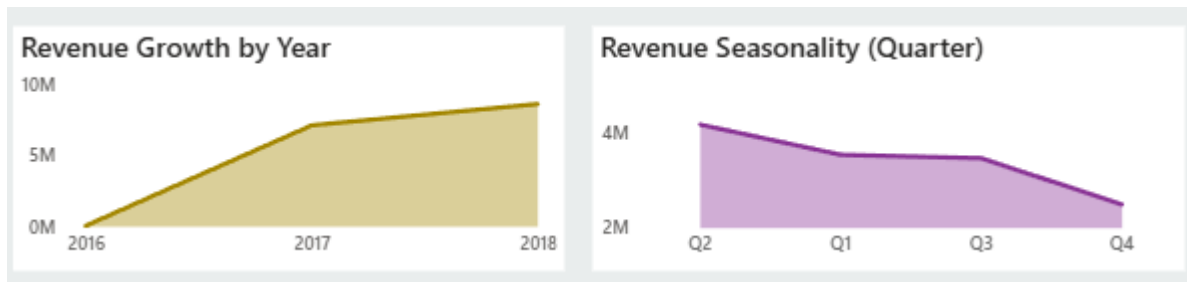
SP drives both revenue and delivery delays, indicating operational pressure in high-demand regions.

6 Revenue Growth & Seasonality

Question Statement

How has revenue grown annually, and are there seasonal patterns?

Visualization



Explanation

Yearly Trend:

- Revenue shows consistent growth from 2016 to 2018.
- Indicates strong market expansion.

Quarterly Trend:

- Q2 and Q3 show higher revenue peaks.
- Q4 decline suggests seasonal or promotional pattern changes.

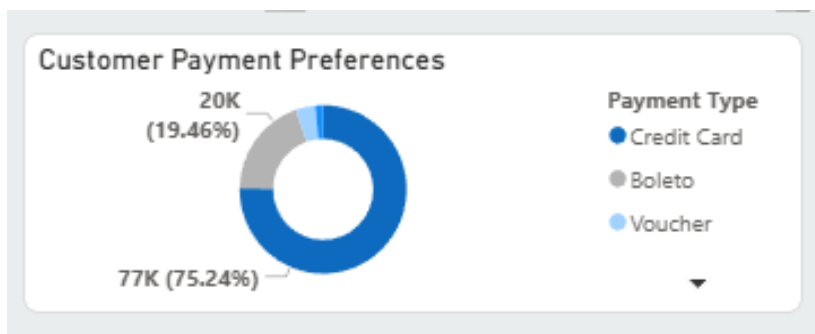
This supports strategic campaign planning.

7 Customer Payment Preferences

Question Statement

What payment methods are preferred by customers?

Visualization



Explanation

- Credit Card accounts for ~75% of transactions.
- Boleto and Voucher represent smaller segments.

Business Impact:

- Credit card partnerships are critical.
 - Risk exposure tied to card transaction fees.
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8 Customer Satisfaction Analysis

Question Statement

Which product categories receive the highest and lowest ratings?

Visualization



Explanation

Highest Rated:

- CDs & DVDs
- Fashion (Children)
- Books

Lowest Rated:

- Furniture-related categories
- Large household goods

Insight:

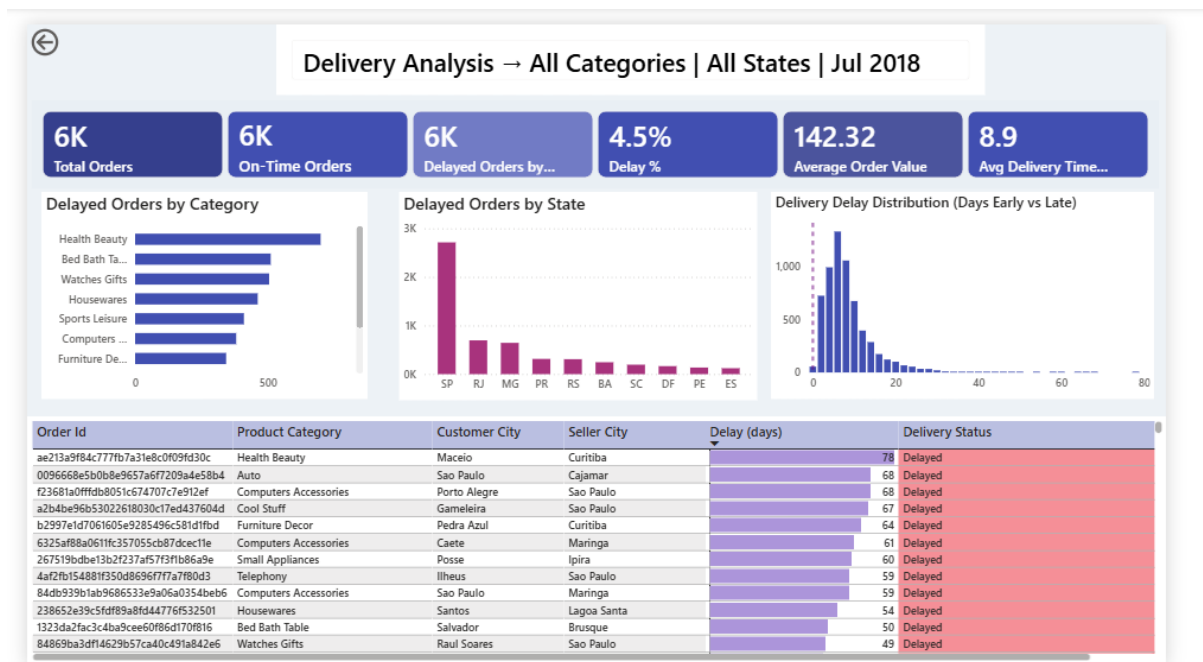
Customer dissatisfaction aligns with longer delivery times in bulky product segments.

9 Delivery Delay Investigation (Drillthrough Page)

Question Statement

What are the root causes of delayed deliveries?

Visualization



Include:

- Late Orders by Category
- Late Orders by State
- Delivery Time Distribution
- Detailed order table

Explanation

This page allows granular analysis of late deliveries.

Key Observations:

- Health & Beauty and Furniture show higher late counts.
- SP state has the highest number of delayed orders.
- Delivery delay distribution is right-skewed, indicating a few extreme outliers.

The order-level table helps identify high-delay cases (50+ days), supporting operational case review.

1 0 Overall Business Conclusion

Revenue growth is strong (17% YoY), but delivery time remains relatively high at 12.5 days.

Operational delays are concentrated in:

- Furniture & bulky categories
- High-demand states (SP)

While SLA compliance is acceptable (8% late rate), improving fulfillment efficiency can significantly enhance customer experience.

Recommendations

1. Optimize logistics network for heavy product categories.
 2. Introduce regional warehouses for SP and high-volume states.
 3. Implement predictive delivery risk monitoring.
 4. Improve last-mile tracking integration.
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Technical Implementation Summary

- Data modeling using star schema
 - Time intelligence with Calendar table
 - YoY calculation using DAX
 - Delivery time computed using DATEDIFF
 - Drillthrough-enabled investigation page
 - Conditional formatting for KPI emphasis
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Tools Used

- Microsoft Power BI
 - DAX (Data Analysis Expressions)
 - Data Modeling (Relationships)
 - Time Intelligence Functions
 - Conditional Formatting
 - Drillthrough Analysis
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Interactive KPI Tooltip Analysis

Question Statement

How can users quickly understand detailed performance metrics without navigating away from the dashboard?

Visualization

(Screenshot: KPI Tooltip popup showing Total Revenue, Total Orders, Avg Rating, Delay %, Average Order Value)



Explanation

To enhance interactivity and user experience, a custom tooltip page was created in Power BI. When a user hovers over key visuals, a contextual performance summary appears displaying critical KPIs.

Metrics Included in Tooltip

- **Total Revenue:** 106M
- **Total Orders:** 99K
- **Average Product Rating:** 4.01
- **Delay Percentage:** 8.1%
- **Average Order Value:** 9.17

Purpose of Tooltip

Instead of navigating multiple pages, decision-makers can instantly access deeper metrics while analyzing any chart.

This improves:

- Dashboard usability
- Analysis speed
- Executive decision efficiency

Business Value

The tooltip enables stakeholders to:

- Evaluate customer satisfaction while viewing revenue
- Compare quality vs performance simultaneously

- Identify risk areas without switching pages

Example:

While analyzing a slow category, users can immediately check if it also has poor ratings or low order value.

Technical Implementation

The tooltip was implemented using a dedicated tooltip page:

1. Created a new report page
2. Set page type → Tooltip
3. Designed KPI cards layout
4. Assigned tooltip page to visuals
5. Enabled hover interaction

This approach ensures reusable contextual insights across the dashboard.

Insight Supported

The tooltip helps correlate:

- Delivery performance
- Customer satisfaction
- Revenue quality

Providing multidimensional analysis in a single interaction.